

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Revision Test 2026

Physics — Class 9 (Science Group)

Syllabus: Unit 1 — Physical Quantities & Measurement | Unit 2 — Kinematics | Unit 3 — Dynamics-I | Unit 4 — Dynamics-II | Unit 5 — Pressure & Deformation in Solids | Unit 6 — Work & Energy (till 6.4) | Unit 9 — Nature of Science and Physics

Total Marks: 31

Time: 1 Hour 30 Minutes

Paper: Exam-13 (Version 1)

1. Attempt **ALL** questions. There are **no choices** in this paper.
2. Section A (12 marks): 12 MCQs. Fill the correct bubble. Suggested time: 15 minutes.
3. Show complete working for every numerical: formula, substitution, answer with correct units.
4. Use black or blue ink only. Pencil is allowed only for diagrams. Calculators are permitted.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

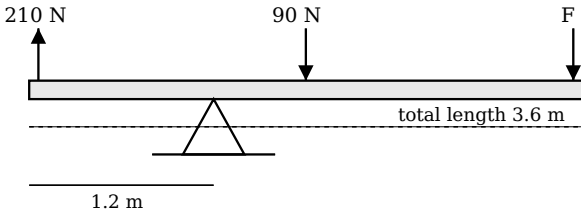
#	Q.1 Choose the best possible option.	A	B	C	D	Answer
1	The most reasonable estimate for the height of a classroom door is:	0.2 m	2 m	20 m	200 m	(A)(B)(C)(D)
2	A rectangle measures 4.5 cm by 2.11 cm. Its area, rounded to the correct number of significant figures, is:	9.495 cm ²	9.50 cm ²	9.5 cm ²	10 cm ²	(A)(B)(C)(D)
3	A student measures the same wire five times and records 20.1, 20.3, 20.2, 20.4 and 20.0 cm. The value that should be reported is:	20.0 cm	20.4 cm	20.2 cm	101.0 cm	(A)(B)(C)(D)
4	Nothing in the universe can travel faster than the speed of light in vacuum, which is about:	3×10^5 m/s	3×10^6 m/s	3×10^8 m/s	3×10^{10} m/s	(A)(B)(C)(D)
5	The reading shown by the speedometer of a moving car at one particular instant is its:	average speed	instantaneous speed	uniform velocity	acceleration	(A)(B)(C)(D)
6	Which laboratory instrument measures the weight of a body directly?	Metre rule	Force meter (spring balance)	Measuring cylinder	Stopwatch	(A)(B)(C)(D)
7	Which one of the following is a non-contact force?	Friction	Tension in a rope	Magnetic force	Air resistance	(A)(B)(C)(D)
8	The second condition of equilibrium requires that:	the sum of all forces is zero	the sum of all torques is zero	the velocity is zero	the acceleration is constant	(A)(B)(C)(D)
9	The line along which a force acts is called its:	axis of rotation	line of action	moment arm	centre of gravity	(A)(B)(C)(D)
10	The pressure exerted by a liquid at a certain depth does NOT depend on:	the depth	the density of the liquid	the cross-sectional area of the container	the value of g	(A)(B)(C)(D)
11	Solar photovoltaic cells convert sunlight directly into electricity. They are mainly made from the semiconductor:	copper	silicon	aluminium	graphite	(A)(B)(C)(D)
12	The branch of physics that studies the fundamental particles making up matter and the forces between them is:	Plasma Physics	Particle Physics	Thermal Physics	Geophysics	(A)(B)(C)(D)

SECTION B — Short Questions (3 × 3 = 9 Marks)

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Q.2	Attempt all three parts. Show your working.	Marks
(i)	A swimmer heads due north across a river at 3 m/s while the current carries them due east at 4 m/s. Find the magnitude of the swimmer's resultant velocity. State whether velocity is a scalar or a vector quantity and give a reason.	3
(ii)	A bus starts from rest and accelerates uniformly to 20 m/s in 10 s. It then travels at this constant speed for a further 25 s. Calculate (a) its acceleration during the first stage, and (b) the total distance it covers in the whole 35 s.	3
(iii)	Define gravitational field strength and write its formula. An object of mass 8 kg is taken to a planet where the gravitational field strength is 3.7 N/kg. Calculate its weight there. Would its mass change on that planet? Give a reason.	3

SECTION C — Long Questions (2 × 5 = 10 Marks)

Q.3	Attempt both questions. Show complete working.	Marks
(i)	(a) State the two conditions of equilibrium and write the mathematical form of each. [2] (b) A uniform wooden plank of weight 90 N and length 3.6 m rests on a single support placed 1.2 m from its left-hand end, as shown below. A boy of weight 210 N stands on the left-hand end. Calculate the downward force F that must be applied at the right-hand end to keep the plank balanced. [3] 	5
(ii)	(a) Define power, write its formula and state its SI unit. [1] (b) A crane lifts a steel girder of mass 850 kg through a vertical height of 24 m in 40 s. Taking $g = 10 \text{ m/s}^2$, calculate (i) the gain in potential energy of the girder, and (ii) the power output of the crane in kilowatts. [3] (c) Name the energy conversion that takes place inside the crane's electric motor while it lifts the girder. [1]	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Units 1, 2, 3, 4, 5, 6 (till 6.4) and 9

— End of Question Paper —